

HORNSBY GIRLS' HIGH SCHOOL



2007 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics

Total Marks – 120
Attempt Questions 1-10
All Questions are of equal value

Begin each question on a NEW SHEET of paper, writing your student number and question number at the top of the page. Extra paper is available.

Question 1 (12 marks) Use a SEPARATE sheet of paper.	Marks
(a) State the exact value of $\cos 135^\circ$	2
(b) Solve for x : $2x - 11 \leq 5x + 6$	2
(c) Find the arc length of a sector with central angle 60° and radius 12cm. Leave answer in exact form.	2
(d) Evaluate $\log_3 5$ correct to 2 decimal places	2
(e) Solve for x : $ 2x - 1 = 17$	2
(f) Jack's Jean Junction had a sale on its jeans. Susie paid \$108 for a pair of jeans after a 20% discount. Calculate the original price of her jeans.	2

General Instructions

- Reading Time- 5 minutes
- Working Time – 3 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a fresh sheet of paper.

Total marks (120)

- Attempt Questions 1-10
- All questions are of equal value

Question 2 (12 marks) Use a SEPARATE sheet of paper.

(a) Differentiate the following

(i) $y = 3x^3 - 4x^2 - 8x + 7$

(ii) $y = e^{4x}$

(iii) $y = x \sin x$

(b) Evaluate: $\sum_{k=1}^4 k^3$

(c) Find the limiting sum of the geometric series: $8 - 4 + 2 - 1 \dots$

(d) State the domain and range of the function $y = 3\sqrt{x} + 1$

(e) Show that the centre and radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ is $(2, -3)$ and 5cm respectively.

Marks

1

1

2

2

2

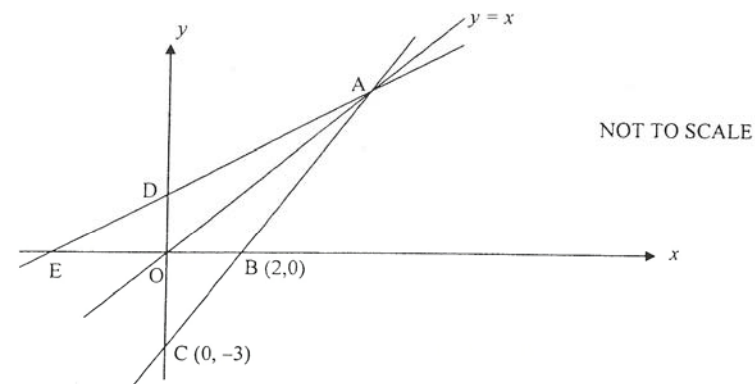
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2

Question 3 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a)



The diagram shows the line $y = x$ and the points $B(2, 0)$ and $C(0, -3)$. The line BC intersects the line $y = x$ at the point A . The points D and E lie on the y -axis and x -axis respectively.

(i) Show that the gradient of BC is $1\frac{1}{2}$

(ii) Find the equation of the line BC .

(iii) Find the coordinates of the point A .

The coordinates of the point E are $(-3, 0)$.

(iv) Show that the equation of the line AE is $2x - 3y + 6 = 0$.

(v) Find the coordinates of the point D .

(vi) Prove that $\triangle AOB \cong \triangle AOD$.

(vii) Find the area of the quadrilateral $OBAD$.

1

2

2

2

1

2

2

Question 4 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) Find:

(i) $\int \left(\frac{4}{x} - \frac{1}{x^2} \right) dx$

2

(ii) $\int_{-1}^1 (3x - 5)^3 dx$

2

(b) If the 4th term and 13th term of an arithmetic sequence are 16 and -2 respectively, find the first term and common difference.

2

(c) For the quadratic equation $3x + 5 - 2x^2 = 0$ find:

(i) $\alpha + \beta$

1

(ii) $\alpha\beta$

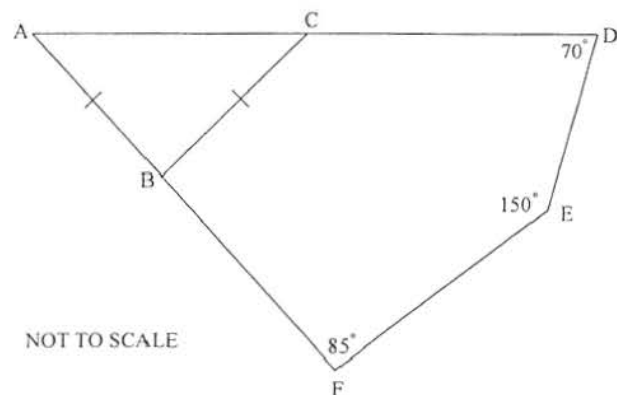
1

(iii) $\alpha^2 + \beta^2$

2

(d) Find the size of $\angle ABC$, giving reasons.

2



Question 5 (12 marks) Use a SEPARATE sheet of paper.

Mark

(a) The displacement of a particle is shown in the table below. The particle comes to rest at $t = 3$. Calculate the distance travelled by the particle in the first 5 seconds.

1

t seconds	0	1	2	3	4	5
x metres	-44	-12	8	23	10	2

(b) If $\log_e a = 4.2$ and $\log_e b = 3$ find:

(i) $\log_e ab$

1

(ii) $\log_e \sqrt{\frac{x}{a}}$

2

(c) Consider the curve given by $y = 2x^3 - 3x^2 - 12x$

(i) Find $\frac{dy}{dx}$

1

(ii) Find the coordinates of the two stationary points and determine their nature

3

(iii) Find the coordinates of the inflexion point

2

(iv) Sketch the curve for $-2 \leq x \leq 3$.

2

Question 6 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) A parabola has equation $y^2 + 8y - 3 = 1 - 20x$

(i) Show that the parabola can be written as $(y + 4)^2 = -20(x - 1)$

2

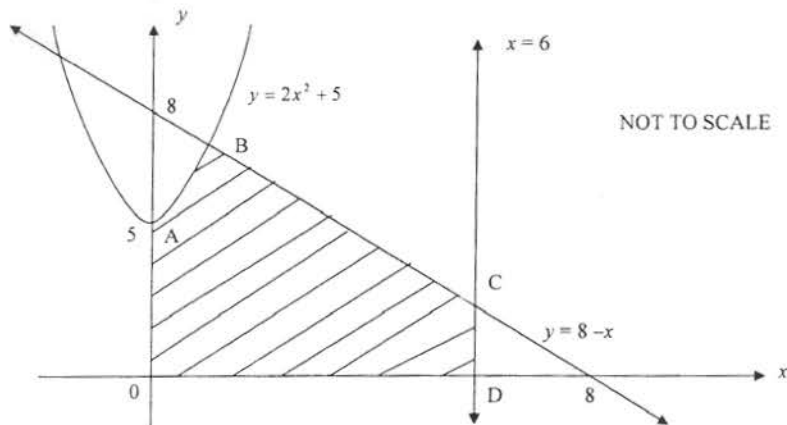
(ii) Find the coordinates of the vertex

1

(iii) Find the coordinates of the focus

1

(b)



In the diagram, the shaded region OABCD is bounded by the curve $y = 2x^2 + 5$, the lines $y = 8 - x$, $x = 6$ and the x and y -axes.

(i) Show that B has coordinates (1, 7)

2

(ii) Calculate the area of the shaded region.
Leave your answer in exact form.

3

(c) A die is rolled twice and the outcomes on the uppermost faces are added together.

(i) Find the probability of getting a score less than 10

2

(ii) If it is certain that a 2 will appear on one of the uppermost faces, find the probability of getting a score less than 5.

1

Question 7 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) Solve for x : $(2^x)^x = 4^{1-x}$

2

(b) Let A be the point (-2, 0) and B be the point (6, 0). At P(x, y), PA meets PB at right angles.

(i) Show that the gradient of PA is $m_1 = \frac{y}{x+2}$

1

(ii) Find an equation for the locus of P

2

(c) Boat A sails 15km for port P on a bearing of 055° .
Boat B sails from P for 25 km on a bearing of 135°

(i) Show the angle $APB = 80^\circ$

1

(ii) Calculate their distance apart to 1 decimal place.

2

(d) The amount, A, of caffeine left in the blood t hours after consuming food or drink containing caffeine is calculated using the formula

$$A = Qe^{-0.17t}$$

where Q is the original quantity of caffeine consumed and t is the time in hours after consuming the caffeine.

(i) Jane likes her coffee strong. At 8am she drank 2 cups of coffee containing a total of 25 mg of caffeine. How much caffeine will be in her blood at 12 noon?
Answer correct to 2 decimal places.

2

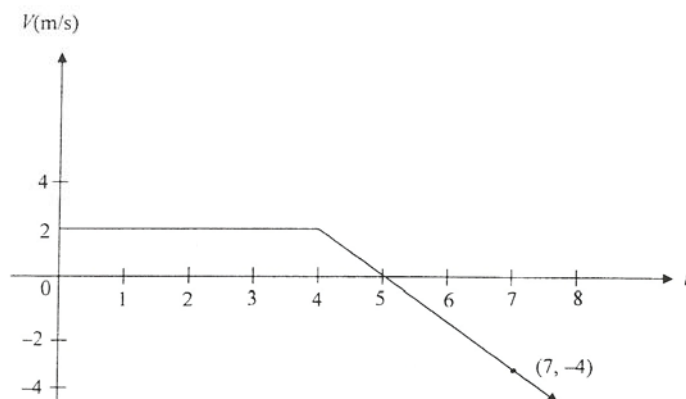
(ii) Calculate the time when 80% of the caffeine she had consumed had been removed from her blood. Answer correct to the nearest minute.

2

Question 8 (12 marks) Use a SEPARATE sheet of paper.

Marks

- (a) The diagram shows the velocity, V /m/s, of a particle moving in a straight line at time t seconds.



- (i) When was the particle stationary?
- (ii) Between what times was the acceleration zero?
- (iii) What was the acceleration after 5 seconds?
- (iv) How far was the particle from its starting position and in what direction was it travelling 7 seconds after the start of the motion?
- (b) (i) Prove the identity $\tan^2 \theta + 1 = \sec^2 \theta$
- (ii) The section of the curve $y = \tan 2x$ which is between $x = \theta$ and $x = \frac{\pi}{8}$ is rotated about the x -axis. Find the exact volume of the solid.

1

2

2

2

2

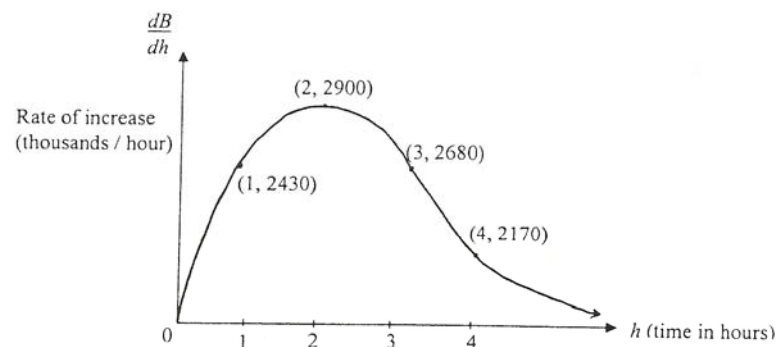
3

Question 9 (12 marks) Use a SEPARATE sheet of paper.

Marks

- (a) The number of bacteria (B) in an untreated infection is increasing. The diagram shows the birth rate of bacteria (h , in thousands per hour).

The equation of the rate of increase is $\frac{dB}{dh} = 4he^{-0.5h}$.



- (i) Use the trapezoidal rule with 4 function values to estimate the total number of bacteria born in the first 3 hours.
- (ii) What value does the rate of increase in the number of bacteria approach as h increases?
- (iii) Initially, the number of bacteria present is 1000. Without integrating, sketch a curve to represent the number of bacteria present in the first h hours of the infection.

2

1

2

- (b) Richard borrowed \$100 000 to buy an apartment, which he is going to rent. He plans to repay \$500 every fortnight from his salary. On every 4th week he will repay an extra \$800 from the rent he receives. Interest on the loan is 6.5% pa reducing fortnightly. Let A_n = the amount owing on the loan after n fortnights and $R = 1.0025$.

- (i) Show that $A_2 = 100000R^2 - 500(R+1) - 800$

1

- (ii) Show that $A_4 = 100000R^4 - 500(R^3 + R^2 + R + 1) - 800(R^2 + 1)$.

2

- (iii) Hence show that A_{100} can be calculated by evaluating:

$$A_{100} = 100000R^{100} - 500 \times \frac{R^{100} - 1}{R - 1} - 800 \times \frac{R^{100} - 1}{R^2 - 1}$$

2

- (iv) What percentage of the money repaid in the first 100 repayments will be interest?

2

Question 10 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) (i) Show that the equation $y = \log_e(x-1)$ can be written as $x = 1 + e^y$

1

(ii) Find the size of the area enclosed by the curve $y = \log_e(x-1)$, the x -axis and the line $x = 5$.

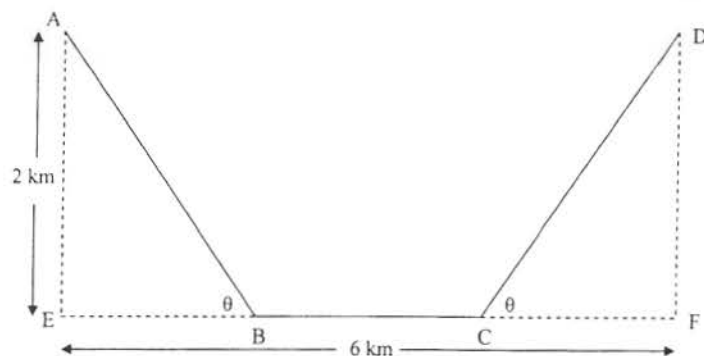
2

(b) Using the quotient rule or otherwise, show that the derivative of

$$\frac{4 - 2\cos\theta}{\sin\theta} \text{ is } \frac{2 - 4\cos\theta}{\sin^2\theta}$$

2

(c)



In the diagram above, A represents Claudia's home and D her school. AEFD is a rectangle where $AE = 2$ km and $EF = 6$ km. Claudia walks to school every morning along the route ABCD. Along AB and CD she walks with a speed of 4 km/hr but along BC she doubles her speed.

(i) Show that the total time needed for Claudia to walk to school, in minutes, is given by:

$$t = 15 \left(3 + \frac{4 - 2\cos\theta}{\sin\theta} \right)$$

4

(ii) Using your answer to part (b) or otherwise, show that the minimum amount of time needed for Claudia to walk to school is approximately 97 minutes.

3

END OF EXAMINATION

12 - TRIAL - 2007 - MATHEMATICS

a) $\cos 135^\circ = -\frac{\sqrt{2}}{2}$

$2x-11 \leq 5x+6$

$-3x \leq 17$

$x \geq -\frac{17}{3}$

$x \geq -5\frac{2}{3}$

$\theta = \frac{\pi}{3}$

$r = 9$

$\ell = \frac{\pi}{3} \times 12$

$\ell = 4\pi \text{ cm}$

$g_5 = \log_3 5$

$x-1=17, 2x-1=-17$

$2x=18, 2x=-16$

$x=9, x=-8$

108 represents 80%

$0.8x = 108$

$x = \$135$

a) i) $y' = 9x^2 - 8x - 8$

$= 4e^{4x}$

$= x \cos x + \sin x$

$k^3 = 1 + 8 + 27 + 64$

$= 100$

$8, r = -\frac{1}{2}$

$\infty = \frac{3}{2}$

$S_{\infty} = 5\frac{1}{3}$

$\text{min: } \{x \in \mathbb{R} \text{ where } x \geq 0\}$

$\text{je: } \{y \in \mathbb{R} \text{ where } y \geq 1\}$

$-4x+4+y^2+6y+9=12+4+9$

$(x-2)^2 + (y+3)^2 = 25$

$\text{centre is } (2, -3), r=5$

Q3) a) i) $m_{BC} = \frac{3}{2} = 1\frac{1}{2}$

ii) $y = \frac{3}{2}x - 3$ or $3x - 2y - 6 = 0$

iii) $3x - 2x - 6 = 0$ $3x - 2y - 6 = 0$

$x - 6 = 0$

$x = 6$

$A(6, 6)$

iv) $(6, 6)$ & $(-3, 0)$

$\frac{y}{x+3} = \frac{6}{9}$

$\frac{y}{x+3} = \frac{2}{3}$

$3y = 2x + 6$

$\therefore \text{equation AE is } 2x - 3y + 6 = 0$

v) For D sub $x=0$

$-3y + 6 = 0$

$y = 2$ $\therefore D \text{ is } (0, 2)$

vi) $\triangle AOB$ & $\triangle AOD$

$-OA \text{ is common}$

$-OB = OD = 2 \text{ units}$

$- \angle AOB = \angle AOD = 45^\circ$ (line $y=x$ makes 45° with the coordinate axes)

$\therefore \triangle AOB \cong \triangle AOD$ (SAS)

vii) Area $\triangle AOB = \frac{1}{2} \times 2 \times 6$

$= 6$

$\therefore \text{Area OBAD} = 2 \times 6 = 12 \text{ u}^2$

Q4) a) i) $\int \frac{4}{x} - x^{-2} dx$

$= 4 \ln x + \frac{1}{x} + c$

ii) $\int_1^4 (3x-5)^3 dx = \left[\frac{(3x-5)^4}{12} \right]_1^4$

$= \frac{(-2)^4}{12} - \frac{(-8)^4}{12}$

$= -340$

Q4b) $a+3b=16$ — (1)

$a+12b=-2$ — (2)

$-9d=18$

$\therefore d=-2$

sub into (1) $a+(3x-2)=16$

$\therefore a=22$

$\therefore \text{first term is } 22 \text{ \& comm diff} = -2$

c) $-2x^2 + 3x - 5 = 0$

i) $\alpha + \beta = \frac{3}{2}$

ii) $\alpha\beta = -\frac{5}{2}$

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= \left(\frac{3}{2}\right)^2 - 2\left(-\frac{5}{2}\right)$
 $= \frac{29}{4}$

d) $\angle A = 55^\circ$ ($\angle \text{sum of quad} = 360^\circ$)

$\angle ACB = 55^\circ$ (base $\angle$ of isos $\triangle$)

$\therefore \angle ABC = 70^\circ$ ($\angle \text{sum of } \triangle = 180^\circ$)

Q5) a) $32 + 20 + 15 + 13 + 8 = 88 \text{ m}$

b) $\log_x ab = \log_x a + \log_x b$
 $= 7.2$

ii) $\log_x \sqrt{\frac{x}{a}} = \frac{1}{2} [\log_x x - \log_x a]$
 $= \frac{1}{2} [1 - 4.2]$
 $= -1.6$

c) $y = 2x^3 - 3x^2 - 12x$

i) $\frac{dy}{dx} = 6x^2 - 6x - 12$

ii) $6x^2 - 6x - 12 = 0$ for stat pts

$x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

$\therefore x=2, x=-1$

$y = -20, 7$

$\frac{d^2y}{dx^2} = 12x - 6$

when $x=2, \frac{d^2y}{dx^2} = 18 > 0 \therefore (2, -20)$ is a min TP

when $x=-1, \frac{d^2y}{dx^2} = -18 < 0 \therefore (-1, 7)$ is a max TP

iii) $y'' = 12x - 6 = 0$ for POI

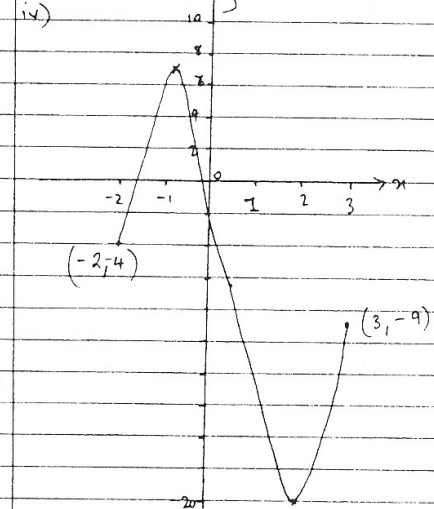
$12x = 6$

$\therefore x = \frac{1}{2}$

$y = -6\frac{1}{2}$

test $0 \frac{1}{2} \frac{1}{6}$

change in concavity
 $\therefore (\frac{1}{2}, -6\frac{1}{2})$ is a P.O.I



when $x = -2, y = -4$

when $x = 3, y = -9$

Q6) a) i) $y^2 + 8y + 16 = -20x + 1 + 3 + 16$

$y^2 + 8y + 16 = -20x + 20$

$(y+4)^2 = -20(x-1)$

ii) vertex $(1, -4)$

iii) $a = 5$

$\therefore \text{focus } (-4, -4)$



$$2x^2 + x - 3 = 0$$

$$(2x+3)(x-1) = 0$$

$$x = -\frac{3}{2}, x = 1$$

$$y = 7$$

B has coord (1, 7)

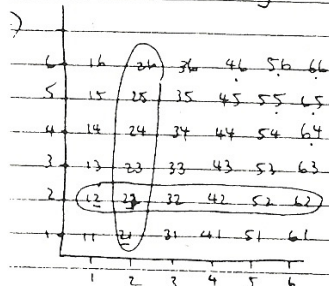
$$i) \text{ Area} = \int_0^1 2x^2 + 5 \, dx + \int_1^6 8 - x \, dx$$

$$= \left[\frac{2x^3}{3} + 5x \right]_0^1 + \left[8x - \frac{x^2}{2} \right]_1^6$$

$$= \left(\frac{2}{3} + 5 \right) + \left((48 - 18) - \left(8 - \frac{1}{2} \right) \right)$$

$$= 5\frac{2}{3} + 30 - 7\frac{1}{2}$$

$$A = 28\frac{1}{6} \text{ u}^2$$



$$P(\text{sum} < 10) = \frac{30}{56} = \frac{5}{6}$$

$$P(\text{sum} < 5) = \frac{3}{11}$$

$$7) a) (2^x)^x = 4^{1-x}$$

$$2x^2 = 2^{2-2x}$$

$$x^2 = 2 - 2x$$

$$x^2 + 2x - 2 = 0$$

$$x = \frac{-2 \pm \sqrt{12}}{2}$$

$$x = -1 \pm \sqrt{3}$$

$$i) P(x, y) \quad A(-2, 0) \quad B(6, 0)$$

$$M_{PA} = \frac{y}{x+2}$$

$$M_{PB} = \frac{y}{x-6}$$

$$\therefore \frac{y}{x+2} = \frac{6-x}{y}$$

$$\therefore y^2 = 6x - x^2 + 12 - 2x$$

$$y^2 = 4x - x^2 + 12$$

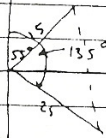
$$\therefore x^2 + y^2 - 4x - 12 = 0$$

$$c) i) \angle APB = 80^\circ$$

$$ii) AB^2 = 15^2 + 25^2 - 2(15)(25) \cos 80^\circ$$

$$AB = 26.8284$$

$$\therefore AB = 26.8 \text{ km (1 dp)}$$



$$d) A = Qe^{-0.17t}$$

$$i) A = 25e^{-0.17(4)}$$

$$A = 12.665$$

$$A = 12.67 \text{ mg}$$

$$ii) \text{ let the original amount} = 100$$

$$20 = 100e^{-0.17t}$$

$$0.2 = e^{-0.17t}$$

$$\ln 0.2 = -0.17t$$

$$\therefore t = \frac{\ln 0.2}{-0.17}$$

$$\therefore t = 9.4672$$

$$t = 9 \text{ hrs } 28 \text{ min (nearest min)}$$

$$\therefore 8 + 9 \text{ hrs } 28 \text{ min} = 5:28 \text{ pm}$$

$$Q8. i) v = 0 \text{ when } t = 5 \text{ sec}$$

$$ii) \text{ acc}^n \text{ was zero between } t = 0 \text{ \& } t = 4 \text{ sec}$$

$$iii) \text{ acc}^n = -\frac{4}{2}$$

$$= -2$$

$$\therefore \text{acc}^n \text{ was } 2 \text{ m/s}^2$$

$$iv) \text{ distance} = \text{area under the curve}$$

$$= \frac{1}{2}(4+5) + \frac{1}{2}(2) \times 4$$

$$= 13 \text{ metres}$$

$$\therefore \text{particle had travelled 13 metres}$$

$$\text{It is travelling in the negative direction after 7 seconds}$$

$$8b) i) \text{ Prove } \tan^2 \theta + 1 = \sec^2 \theta$$

$$\text{LHS} = \tan^2 \theta + 1$$

$$= \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta}$$

$$= \frac{1}{\cos^2 \theta}$$

$$= \sec^2 \theta$$

$$= \text{RHS}$$

$$\therefore \text{LHS} = \text{RHS} \text{ Q.E.D.}$$

$$ii) V = \pi \int_0^{\frac{\pi}{8}} \tan^2 2x \, dx$$

$$= \pi \int_0^{\frac{\pi}{8}} \sec^2 2x - 1 \, dx$$

$$= \pi \left[\frac{1}{2} \tan 2x - x \right]_0^{\frac{\pi}{8}}$$

$$= \pi \left[\left(\frac{1}{2} - \frac{\pi}{8} \right) - 0 \right]$$

$$V = \pi \left[\frac{4 - \pi}{8} \right] \text{ u}^3$$

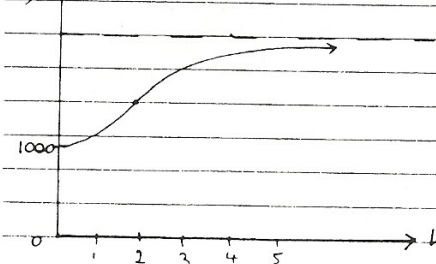
$$Q9. a) i) \text{ Number Bacteria} =$$

$$2 \times (0 + 2680 + 2(2430 + 2900))$$

$$= 6670,000$$

$$ii) \text{ As } h \rightarrow \infty, \frac{dB}{dh} \rightarrow 0$$

$$iii) B(1000s)$$



$$b) \text{ Let } A_n = \text{amount owing after } n \text{ fortnights}$$

$$A_1 = 100000(1.0025) - 500$$

$$A_2 = (100000(1.0025) - 500)(1.0025) - 500 - 800$$

$$\therefore A_2 = 100000(1.0025)^2 - 500(1 + 1.0025) - 800$$

$$\therefore A_2 = 100000R^2 - 500(1+R) - 800$$

$$A_3 = [100000R^2 - 500(1+R) - 800](R) -$$

$$A_3 = 100000R^3 - 500R(1+R) - 800R$$

$$A_3 = 100000R^3 - 500(R^2 + R + 1) - 800R$$

$$A_4 = [100000R^3 - 500(1+R+R^2) - 800R]R -$$

$$= 100000R^4 - 500(R^3 + R^2 + R + 1) - 800R$$

$$A_{100} = 100000R^{100} - 500(1+R+...$$

$$- 800(1+R^2+...+R^{50})$$

$$= 100000R^{100} - 500 \left(\frac{R^{100}-1}{R-1} \right)$$

$$- 800 \left(\frac{R^{50}-1}{R^2-1} \right)$$

$$= 100000R^{100} - 500 \frac{(R^{100}-1)}{R-1} - 800 \frac{(R^{50}-1)}{R^2-1}$$

$$= 100000R^{100} - 500 \frac{(R^{100}-1)}{R-1} - 800 \frac{(R^{50}-1)}{R^2-1}$$

$$iv) A_{100} = 128362.49 - 56724.97 -$$

$$- 45323.328$$

$$A_{100} = \$26314.18$$

$$\text{Total repaid} = 500 \times 100 + 800 \times 50$$

$$= \$90000$$

$$\therefore \text{Interest} = \$16314$$

$$\therefore \text{percentage interest} = 18.1\%$$

$$Q10. a) i) y = \log_e(x-1)$$

$$e^y = x-1$$

$$\therefore x = e^y + 1$$

$$ii) \ln 4$$

$$y = \ln(x-1)$$

$$x = e^y + 1$$

$$y = \ln 4$$

$$x = e^{\ln 4} + 1 = 4 + 1 = 5$$

$$\text{when } x = 5, y = \ln 4$$

$$\therefore \text{Shaded Area} = 5 \ln 4 - \int_1^4 e^y + 1 \, dy$$

$$= 5 \ln 4 - [e^y + y]_1^4$$

$$= 5 \ln 4 - (e^4 + 4 - e - 1)$$

$$= 5 \ln 4 - 4 - \ln 4 + 1$$

$$= 4 \ln 4 - 3$$

$$= 2$$

Q10 (cont.)

$$\begin{aligned}
 \text{b) i) } \frac{d}{dx} \left(\frac{4-2\cos x}{\sin x} \right) &= \frac{\sin x (2\sin x) - (4-2\cos x) \cos x}{\sin^2 x} \\
 &= \frac{2\sin^2 x - 4\cos x + 2\cos^2 x}{\sin^2 x} \\
 &= \frac{2\sin^2 x + 2\cos^2 x - 4\cos x}{\sin^2 x} \\
 &= \frac{2 - 4\cos x}{\sin^2 x} \quad \text{QED!}
 \end{aligned}$$

$$\begin{aligned}
 \therefore t &= 15 \left(3 + \frac{4-2\cos 60}{\sin^2 60} \right) \\
 t &= 96.96 \\
 \therefore t &\approx 97 \text{ minutes}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \sin \theta &= \frac{2}{AB} \\
 \therefore AB &= \frac{2}{\sin \theta} \quad \& \quad CD = \frac{2}{\sin \theta} \\
 \therefore \text{time AB} &= \frac{2}{4\sin \theta} \quad \text{time CD} = \frac{2}{4\sin \theta} \\
 &= \frac{1}{2\sin \theta} \quad = \frac{1}{2\sin \theta}
 \end{aligned}$$

50°	60°	70°
-14.6	0	28.16

$\therefore$ min when $t = 97$

$$\begin{aligned}
 EB &\Rightarrow \tan \theta = \frac{2}{EB} \\
 \therefore EB &= \frac{2}{\tan \theta} \\
 \therefore BC &= 6 - \frac{2}{\tan \theta} - \frac{2}{\tan \theta} \\
 &= 6 - \frac{4}{\tan \theta} \\
 \therefore \text{time BC} &= \frac{6 - \frac{4}{\tan \theta}}{8} = \frac{3\tan \theta - 2}{4\tan \theta} \\
 \therefore t &= \frac{1}{2\sin \theta} + \frac{1}{2\sin \theta} + \frac{3\tan \theta - 2}{4\tan \theta} \\
 &= \frac{1}{\sin \theta} + \frac{3}{4} - \frac{\cos \theta}{2\sin \theta} \\
 &= \left(\frac{3+2-\cos \theta}{4\sin \theta} \right) \times 60 \text{ minutes} \\
 &= \frac{45 + 60 - 30\cos \theta}{\sin \theta} \\
 t &= 15 \left(3 + \frac{4-2\cos \theta}{\sin \theta} \right) \quad \text{QED!}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } t' &= 15 \left(\frac{2-4\cos \theta}{\sin^2 \theta} \right) = 0 \text{ for min time} \\
 t' &= \frac{2-4\cos \theta}{\sin^2 \theta} = 0 \quad \therefore \theta = 60^\circ \\
 2 &= 4\cos \theta \\
 \therefore \cos \theta &= \frac{1}{2}
 \end{aligned}$$